Global Air Cargo Volume 1970 – 2017

Time Series Analysis and Forecast

MIS 748: Time Series Analysis for Business Forecasting

San Diego State University

Spring 2025

Micha Goll

Content

	Executive Summary	**3**
1.	Data Exploration and Preparation	**4**
2.	Results and Performance	**8**
2.1	ACF and PACF Diagnostics	**8**
2.2	Model Evaluation	**9**
2.3	Model Refinement and Selection	**11**
3.	One-Step Forecast for 2018	**13**
4.	Limitations	**14**
5.	Summary and Conclusion	**15**
6.	References	**16**

Executive Summary

This project analyzes and forecasts global air freight patterns based on 1970 - 2017 yearly data from World Bank (2023). The data shows total air cargo volume or, to be more precise, weight in million ton-kilometers and reflects the performance of global trade and logistics. Air freight plays a key role in transporting time sensitive and high value goods between countries and industries. Trends in air cargo help us understand world trade flows, global supply chains structures and general worldwide economic activity. Due to the mentioned economic importance of air freight in global supply chains, accurate forecasting is a powerful tool (IATA, n.d.).

The original data demonstrated a clear trend (upward) and increasing variance over time, which indicated non-stationarity. The variance was stabilized by log transformation, followed by taking the first difference to remove the non-constant mean. The Augmented Dickey-Fuller test confirmed that the transformed series was stationary. Initial identification with the ACF, PACF, and EACF suggested an ARIMA(0,1,1) structure but, upon subsequent comparison of a number of specifications of ARIMA, based on AIC, BIC, and Ljung-Box diagnostic tests on residuals, the ultimate forecasting method of choice was an ARIMA(2,1,1).

Using the model trained on the 1970-2017 data, a one-step ahead forecast for 2018 was conducted. It predicted 1,683,722 million ton-kilometers versus an observed value of 1,673,174. Hence, the forecast error was just 0.63%. The 95% prediction interval for the year 2018 was between 1,552,102 and 1,826,503. This suggests that the model represented the underlying structure of the data in a fitting way and adequately generalized the unobserved.

While the forecast was accurate, one major limitation of the analysis is dependence on annual data. It constrains temporal granularity and excludes modeling intra-year dynamics or seasonality. The relatively small number of observations (48 training points) also constrained model complexity. Future research would benefit from monthly or quarterly data, which would facilitate more precise temporal modeling, enhanced short-run shock identification, and the application of more advanced time series methods (Hyndman & Athanasopoulos, 2018). Integrating external variables like the oil price or trade policy shifts may be added as a helpful component in a larger scale study to assess correlations of air cargo volume with economic variables, or to predict other variables with an air cargo volume forecast.

1. Data Exploration and Preparation

Data utilized for the present analysis is drawn from the World Bank indicator international air freight (million ton-km), which quantifies the volume of goods transported by air freight on a global scale per year (World Bank, 2023). Although the data series begins in the 1960s, problems and inconsistencies in country reporting prior to 1970 necessitated the inclusion of only annual observations from 1970 through 2017. Every record is the total worldwide freight tonnage for a specific calendar year, expressed in millions of ton-kilometers. Data was used in CSV format from the World Bank online database and then processed it through R.

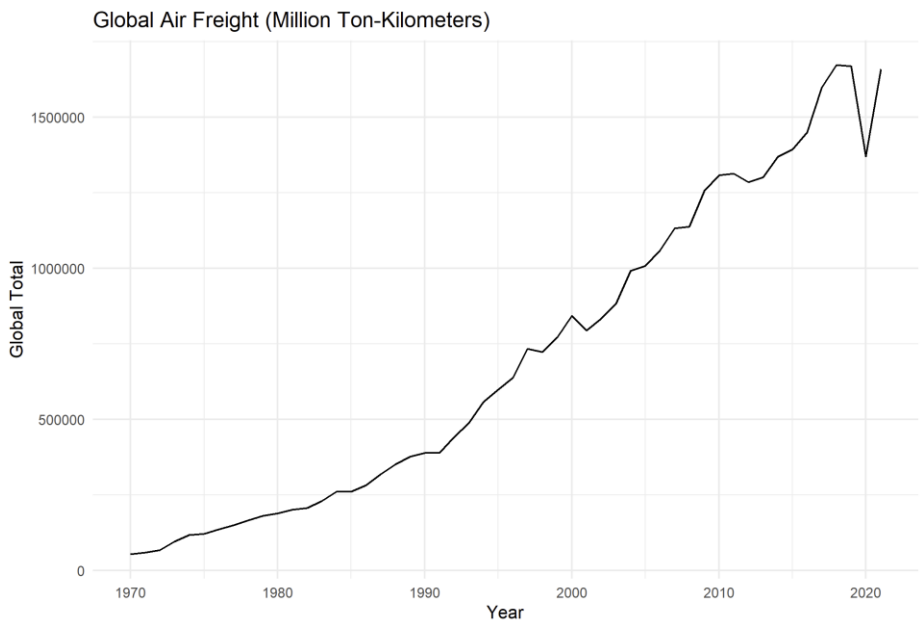

Figure 1. *Annual global air freight volume, 1970–2023 (million ton-kilometers)*

On the subsequent page, figure 2 shows the raw time series data from 1970 to 2017. Although data for 2018 and beyond was available, all data points from 2017 onwards were excluded from the analysis and model training. This was done to prevent distortions from the steep decline in air freight volume as a result of the COVID-19 pandemic, which would have represented an extraneous disruption unrelated to the underlying structural trend. Omitting the pandemic period allowed a cleaner, more homogeneous model to be constructed using only pre-COVID trends. Additionally, this setup allowed for a one-step-ahead forecast for 2018 which enables for direct comparison to actual 2018 values. The goal was to assess out-of-sample predictive ability and model performance.

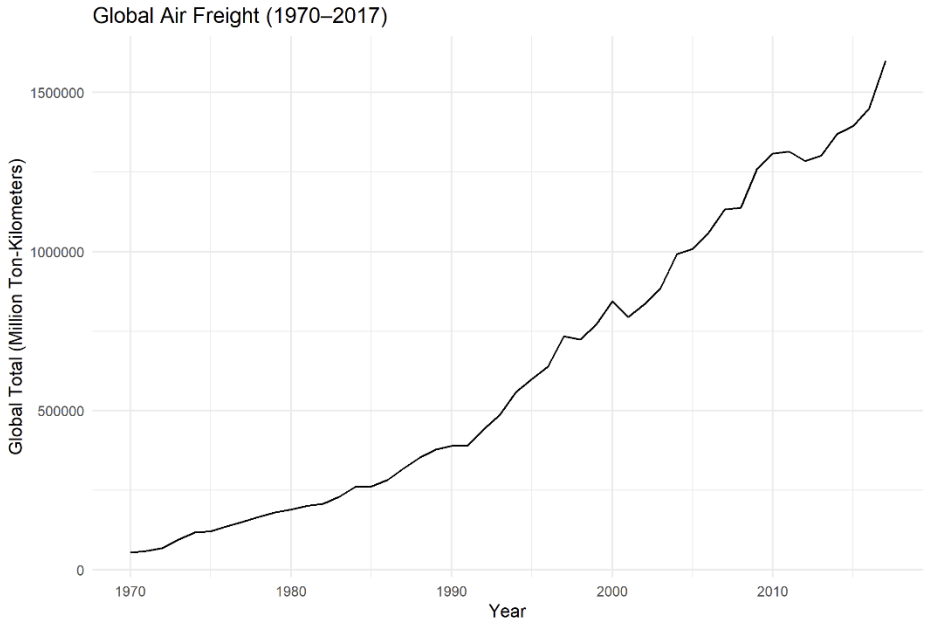

Figure 2. *Annual global air freight volume, 1970–2017 (million ton-kilometers)*

Before modeling, basic summary statistics were computed for the raw series from 1970 to 2017. As shown in table 1, the global air freight volume had a mean of 635,972 and a standard deviation of 468,827, with a minimum value of 53,879 and a maximum of 1,599,661 million ton-kilometers. The wide spread and high variance of the data (variance = 219,798,979,574) suggested substantial heteroskedasticity, supporting the decision to apply a logarithmic transformation to stabilize the variance.

Statistic	Value
Mean	635,972
Median	522,879
Minimum	53,879
Maximum	1,599,661
Variance	219,798,979,574
Std. Dev.	468,827

Table 1. *Summary statistics of global air freight volume (1970–2017)*

Visual examination of the raw data revealed a trend along with growing spread, indicating non-stationarity in both mean and variance. This trend is in line with macroeconomic trends and with exogenous expectations of growth in international trade and logistics capacity over the last half-century (IATA, n.d.).

In order to correct the non-constant variance, a logarithmic transformation was performed on the data. Also, the first difference of the log-transformed values was computed to eliminate the non-constant mean (trend) and attain stationarity. The procedure yielded a differenced log series fluctuating around a constant mean.

Since both logarithmic transformation and first order differencing had already been applied to the original series, the next step was to formally test whether these adjustments were sufficient to achieve stationarity. For this purpose, the Augmented Dickey-Fuller (ADF) test was used (Table 2-4). The test confirmed that the raw and log-transformed series were still non-stationary, with p-values of 0.8359 and 0.7591, respectively, both well above the 0.05 threshold. Only after differencing the log-transformed series did the ADF test produce a p-value below 0.05 and even 0.01, indicating that the transformation was adequate. The resulting series, plotted in figure 3, fluctuates around a constant mean and satisfies the assumptions required for ARIMA modeling.

```
Augmented Dickey-Fuller Test

data:  raw_series
Dickey-Fuller = -1.3446, Lag order = 3,
p-value = 0.8359
alternative hypothesis: stationary
```

Table 2. *ADF test results for raw air freight series (1970–2017). The p-value of 0.8359 indicates non-stationarity*.

```
Augmented Dickey-Fuller Test

data:  log_series
Dickey-Fuller = -1.5369, Lag order = 3,
p-value = 0.7591
alternative hypothesis: stationary
```

Table 3. *ADF test results for log-transformed series. The p-value remains high (0.7591), confirming that log transformation alone does not achieve stationarity.*

```
Augmented Dickey-Fuller Test

data:  diff_series
Dickey-Fuller = -5.0416, Lag order = 3,
p-value = 0.01
alternative hypothesis: stationary
```

Table 4. *ADF test results after applying both log transformation and first differencing. The p-value of 0.01 confirms stationarity was achieved.*

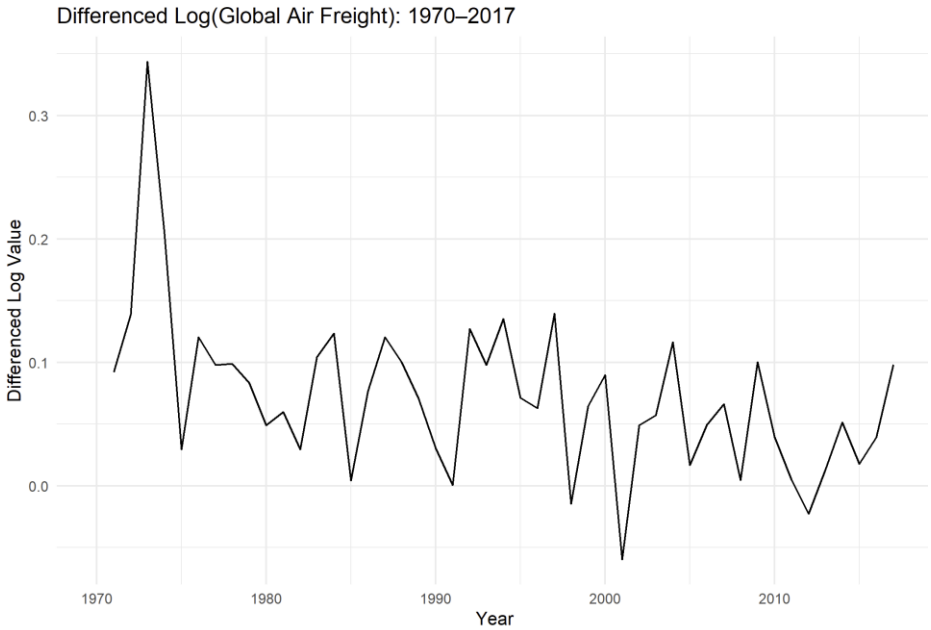

Figure 3. *Log-transformed and differenced air freight volume, 1971–2018*

2. Results and Performance

2.1 ACF and PACF Diagnostics

Once the series had been transformed and confirmed to be stationary, I turned to identifying an appropriate ARIMA model. To guide this step, I examined the ACF and PACF plots of the differenced log series (figure 4). These diagnostic plots offer a visual sense of the time-dependent structure that remains after transformation and are often used to form initial hypotheses for ARIMA model parameters.

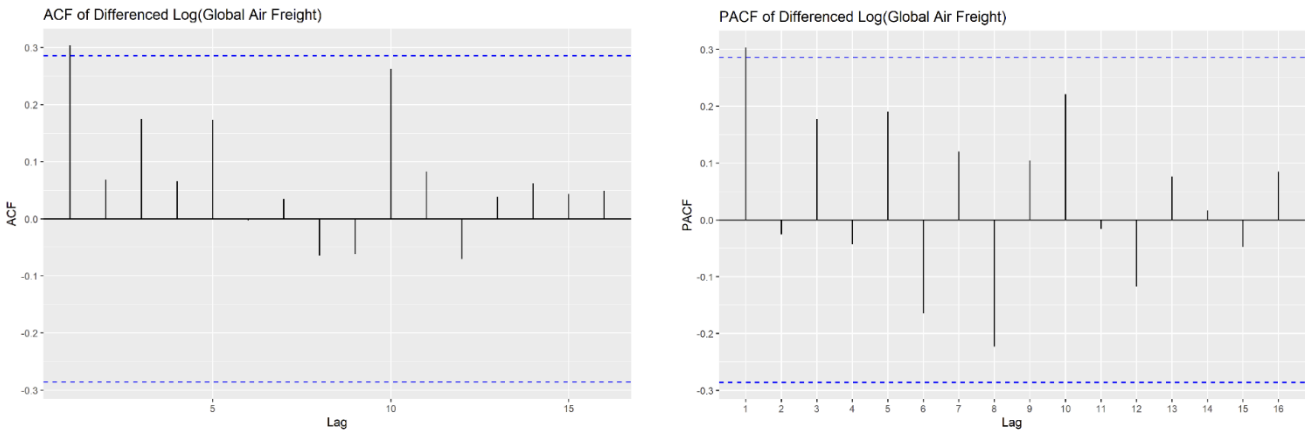

Figure 4. *ACF and PACF plots of differenced log(global air freight), 1971–2017*

The ACF plot showed a decline but not an abrupt one following the first lag, so an MA(1) component could not yet confidently deducted. The PACF declined slowly, indicating possible autoregressive characteristics. I proceeded with model selection based on these insights and tested several candidate models featuring different combinations of AR and MA terms.

2.2. Model Evaluation

Due to the mixed signals from the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), I decided to test ARIMA models that included various combinations of autoregressive (AR) and moving average (MA) components rather than committing to a single model form prematurely. The initial set of models included ARIMA(1,1,0), ARIMA(0,1,1), and ARIMA(1,1,1), chosen to examine whether residual patterns preferred an autoregressive strategy, a moving average strategy, or a blend of both components.

To identify the most suitable model, I cross-compared these specifications on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). ARIMA(1,1,0) had the lowest AIC (-111.42) and BIC (-107.72), which suggested that it achieved the best balance between fit and parsimony. Although ARIMA(0,1,1) and ARIMA(1,1,1) were not poor performers, their higher information criteria suggested they were less preferable.

However, residual behavior also determines model sufficiency. Visual inspection of the ARIMA(1,1,0) residuals revealed no obvious patterns or variance inflation. Nevertheless, the Ljung-Box test did provide a p-value of 0.0124, since residual autocorrelation still existed at some lags. The finding suggested the model might not fully capture the time-dependent character of the data.

ARIMA (1,1,0)
AIC: -111.42
BIC: -107.72

ARIMA (0,0,1)
AIC: -103.42
BIC: -99.72

ARIMA (0,1,1)
AIC: -108.38
BIC: -102.83

Table 5*. AIC, BIC, and Ljung-Box results for the first ARIMA model candidates*

To determine the suitability of the initially favored ARIMA(1,1,0) model, I performed both graphical and statistical residual analysis. The plot of the residuals versus time showed some stability with random fluctuations around zero without apparent trends or variance changes, which graphically indicated properties typical of white noise. Likewise, the ACF plot of

residuals showed that all but one of the autocorrelations fell within the 95% confidence interval, though a noticeable peak at lag 11 crossed the threshold.

Yet, I conducted a more formal test using the Ljung-Box test on the first 10 residual lags. The test yielded a Chi-squared statistic of 22.582 and a p-value of 0.0124. As this finding is below the standard 0.05 criterion, I rejected the null that the residuals are independently distributed. This told me that the model was leaving some autocorrelation uncaptured and that ARIMA(1,1,0) might not be entirely suitable to model the underlying time-dependent structure.

In light of these results, I conducted an exploration of other candidate models for enhancing residual behavior, namely those specifications that would decrease the lingering autocorrelation without making the model too complicated.

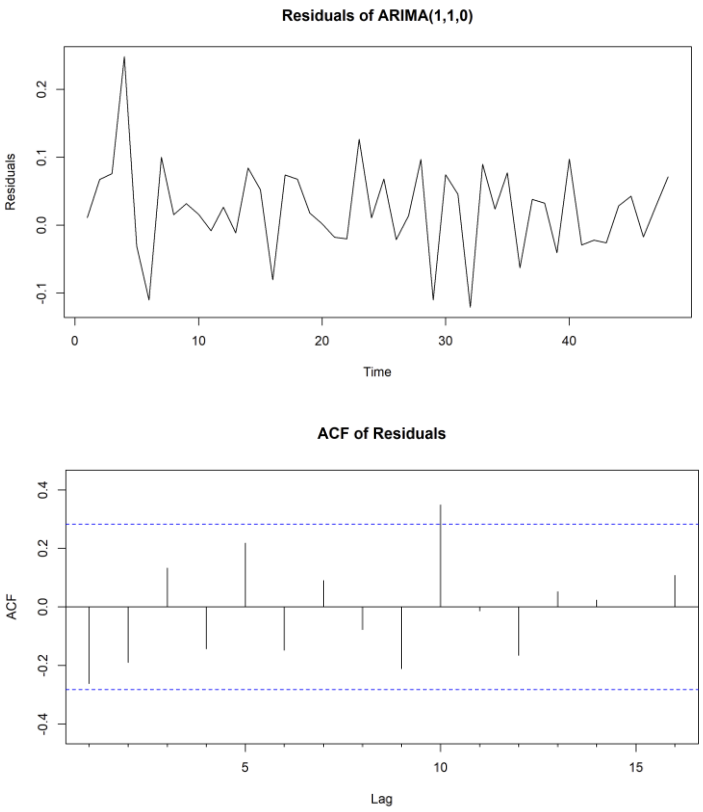

Figure 6. *ARIMA(1,1,0) residuals and ACF of residuals. The residuals fluctuate around zero with no visible trend, whereas the ACF shows a significant spike at lag 11, indicating potential autocorrelation*

```
Box-Ljung test

data:  residuals_010
X-squared = 22.582, df = 10,
p-value = 0.0124
```

Table 6. *Ljung-Box test results for residuals of the ARIMA(1,1,0) model.*

2.3 Model Refinement and Selection

Since residual diagnostics of the ARIMA(1,1,0) model indicated autocorrelation, I proceeded to try out several other models to achieve better performance. I applied ARIMA(1,1,1), ARIMA(2,1,0), and ARIMA(2,1,1) on the log-transformed training data and compared their fit based on AIC, BIC, and residual plots. Of these, ARIMA(2,1,1) looked promising. It also possessed the lowest AIC (-118.88) and BIC (-111.48), but more importantly, its Ljung-Box test p-value was 0.303—well above the 0.05 level, indicating there was no indication of residual autocorrelation.

Before testing the models, I used the Extended Autocorrelation Function (EACF) on the differenced log transformed time series. The result indicated ARIMA(2,1,1) as a good fit (see figure 5).

```
AR/MA
  0 1 2 3 4 5 6 7 8 9 10 11 12 13
0 x o o o o o o o o o o  o  o  o
1 o o o o o o o o o o o  o  o  o
2 o x o o o o o o o o o  o  o  o
3 o x o o o o o o o o o  o  o  o
4 o x o o o o o o o o o  o  o  o
5 x o x o o o o o o o o  o  o  o
6 x o o o o o o o o o o  o  o  o
7 o x o o o o o o o o o  o  o  o
```

Figure 5. *EACF matrix for the differenced log-transformed series*

The other models did possess statistically significant residual structure. These results rendered ARIMA(2,1,1) the best fitting model, with both great sample fit and residual properties consistent with assumptions required for valid forecasting.

```
Model: ARIMA(1,1,0)
AIC: -111.4156
BIC: -107.7153
Ljung-Box p-value: 0.01239821

Model: ARIMA(1,1,1)
AIC: -108.3824
BIC: -102.8319
Ljung-Box p-value: 0.007523122

Model: ARIMA(2,1,0)
AIC: -111.3883
BIC: -105.8378
Ljung-Box p-value: 0.04107524

Model: ARIMA(2,1,1)
AIC: -118.884
BIC: -111.4834
Ljung-Box p-value: 0.3029403
```

Table 7. *Model summary output showing AIC, BIC, and Ljung-Box p-values for four ARIMA specifications evaluated in R.*

Model	AIC	BIC	Ljung-value
ARIMA(1,1,0)	-111.42	-107.72	0.0124
ARIMA(1,1,1)	-108.38	-102.83	0.0075
ARIMA(2,1,0)	-111.39	-105.84	0.0411
ARIMA(2,1,1)	**-118.88**	**-111.48**	**0.303**

Table 8. *Comparison table of ARIMA model performance metrics. ARIMA(2,1,1) achieves the best fit and passes the residual independence test*

As a next step, I verified the model fit of ARIMA(2,1,1). I evaluated the residuals graphically and tested it statistically. The residual series showed no patterns or changes in variance. It fluctuated around zero. Also, the ACF of the residuals showed no significant spikes. All were within the confidence interval. Thus, no autocorrelation exists. The model captured the data structure well.

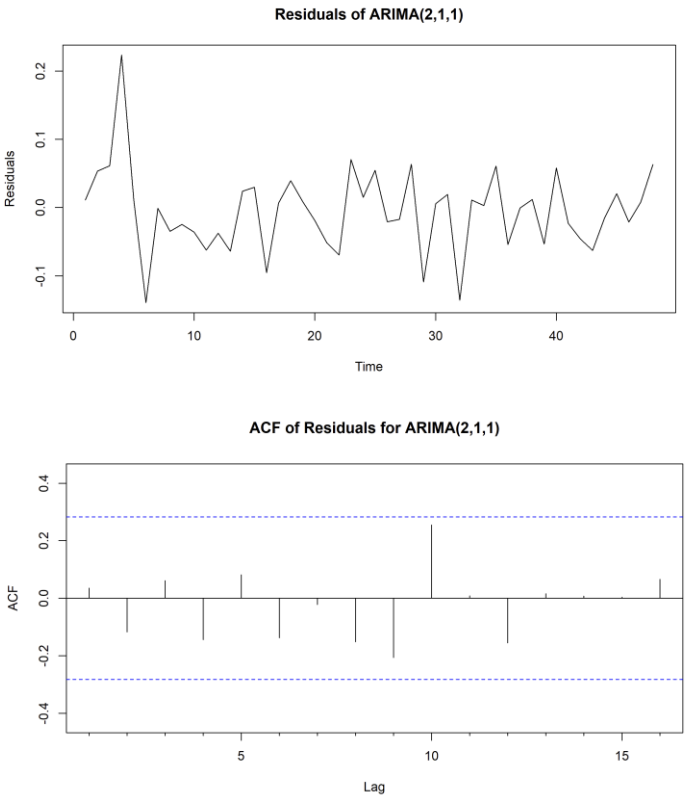

Figure 7. *Residual and ACF plots for the ARIMA(2,1,1) model (1971–2017)*

3. One-Step Forecast for 2018

To estimate the predictive performance of the final model, a one-step-ahead rolling forecast for 2018 was conducted. The ARIMA(2,1,1) model fitted using data for the period 1970-2017 generated an estimated total global air freight volume of 1,683,722 million ton-kilometers. This estimate was very much in agreement with the actual observed value of 1,673,174, having only an absolute error of 10,548. As a percentage of the actual, the forecast error was just 0.63%. A deviation this low is very much a testimonial to the strength and ability of the model to generalize beyond the sample used for training. Results like these suggest that not only is the ARIMA(2,1,1) specification a good fit to previous trends, but one that is also able to make precise short-term predictions in practice.

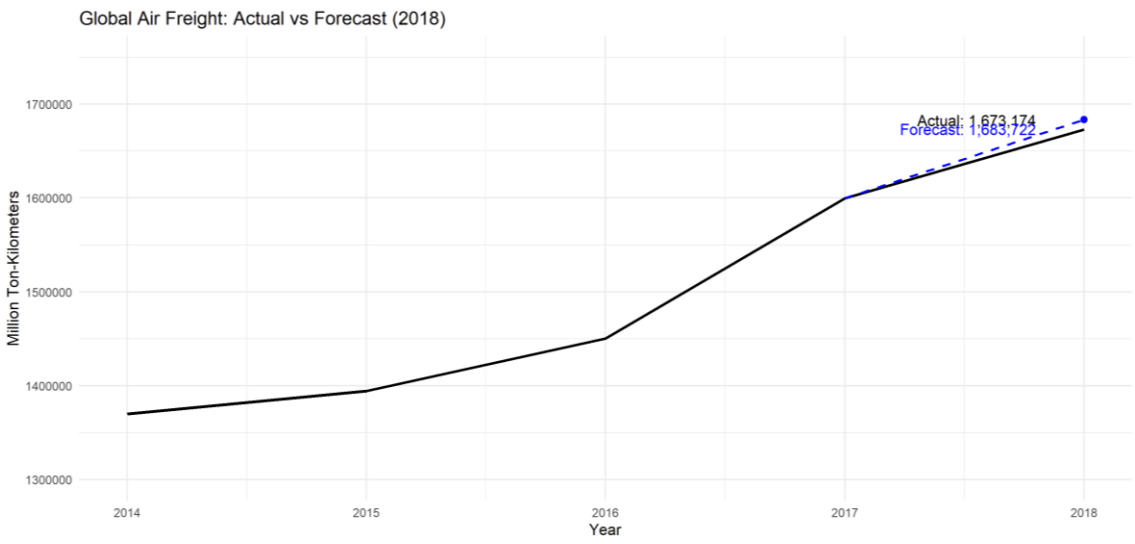

Figure 8. Actual vs. forecasted global air freight for 2018 using the ARIMA(2,1,1) model

4. Limitations

While the ARIMA(2,1,1) model demonstrated strong in-sample performance and yielded an accurate one-step forecast for 2018, several limitations should be noted. Most significantly, the analysis is based on only 48 annual data points (1970–2017), which restricts the use of more advanced forecasting techniques. Models such as SARIMA (Seasonal ARIMA), exponential smoothing state space models (ETS), machine learning approaches like random forests or support vector regression, and deep learning models such as LSTMs typically require more granular and extensive time series data to train effectively and avoid overfitting. Additionally, the annual frequency limits the ability to model short-term dynamics or seasonality, which may be relevant in global air freight patterns. The assumption of stationarity and historical structural stability may not hold in the presence of economic shocks, pandemics, or geopolitical disruptions. Finally, model validation was limited to a one-year forecast horizon, so the model's ability to generalize beyond 2018 remains untested.

5. Summary and Conclusion

This project set out to build a sound forecasting model of worldwide air cargo volume using annual observations between 1970 and 2017. By excluding data from 2019 and onwards, the COVID-19 structural break could be avoided for the data to better suit model training.

A combination of adjustments was used to stabilize the series, including differencing and logarithmic transformation, and the Augmented Dickey-Fuller test confirmed stationarity of the transformed data.

Step-by-step model selection was attained, beginning with ACF and PACF plots analysis and subsequent comparison. Diagnostic results indicated the initially promising ARIMA(1,1,0) model failed to account for autocorrelation, according to the Ljung-Box test. Additional tuning revealed the ARIMA(2,1,1) model to not only have the smallest AIC and BIC but also pass all residual tests.

The one-step-ahead forecast for 2018 yielded a prediction of 1,683,722 million ton-kilometers, only deviating by 0.63% from the actual 2018 value. A testimony to the very high forecasting quality of the model.

While these results are encouraging, the study is limited by the small number of data points, which would not allow for the application of more advanced models such as state-space models, dynamic regression, or machine learning models. Further, using annual data reduces the sensitivity of the model to short term trends and makes it impossible to detect seasonality. Nonetheless, ARIMA(2,1,1) does provide a statistically sound and reasonable model for describing and forecasting global air cargo transports. More advanced studies need to be conducted to assess monthly or quarterly data and fit more advanced models. Also, external factors like the oil price or trade policy changes may be included in a study to assess correlation between air cargo volume and economic and/ or political shifts.

6. References

World Bank. (2023). Air transport, freight (million ton-km) [Data set]. https://data.worldbank.org/indicator/IS.AIR.GOOD.MT.K1

Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: Principles and practice* (2nd ed.). OTexts. https://otexts.com/fpp2/

International Air Transport Association. (n.d.). *Air cargo and global trade*. https://www.iata.org/en/publications/economics/